FIGURE 2 · CONTAINER AND CODE VIEW · AS BUILT, OCTOBER 2026 (PHASE 4.5)

One codebase, two processes, one database: the system as it runs today

What runs on Google sources and Gemini today. Arrows show data flow and name the protocol or payload; a double-headed arrow means both directions.

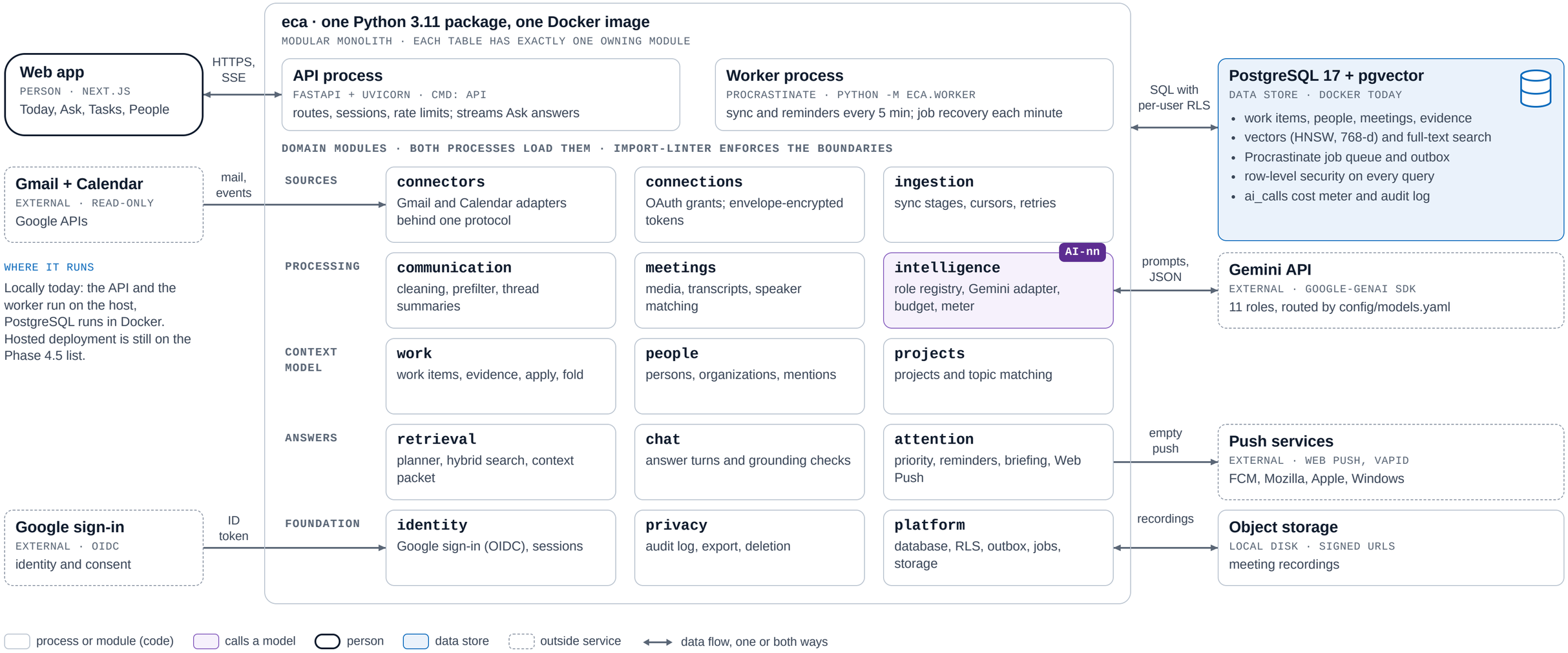


Figure 2. The system as it runs today. One Python package, **eca**, ships as one Docker image and runs as two processes. The **API** serves the Next.js web app and streams answers; the **worker** runs scheduled and queued jobs from a queue that lives in PostgreSQL, so there is no separate message broker. Fifteen domain modules sit behind import boundaries that CI enforces, and every table has exactly one module that writes it. **PostgreSQL 17 with pgvector** is the only stateful service: the context model, vectors, full-text search, the job queue and the outbox share one database under per-user row-level security (RLS). Gemini is reached only through the **intelligence** module, so moving to Azure OpenAI is an adapter change. Reminder pushes carry no content: the browser fetches it after sign-in.